

EXPLORATORY DATA ANALYSIS

SLOT = E2 + TE2

BASIC PYTHON PROGRAMS

1. **print**("Hello, World!")
name = "Aarav"
print("My name is", name)

OUTPUT =

```

Hello, World!
My name is Arav
PS C:\Users\Tinu Goyal\OneDrive\Desktop\Tinu> & "C:\Users\Tinu Goyal\AppData\Local\Programs\Python\Python311\python.exe" "C:\Users\Tinu Goyal\OneDrive\Desktop\Tinu\PRATICE-PY"

```

- ```
2. a = float(input("Enter first number: "))
 b = float(input("Enter second number: "))
```

```
sum_val = a + b
diff = a - b
prod = a * b
quot = a / b if b != 0 else float('inf')
```

```
print("Sum:", sum_val)
print("Difference:", diff)
print("Product:", prod)
print("Quotient:", quot)
```

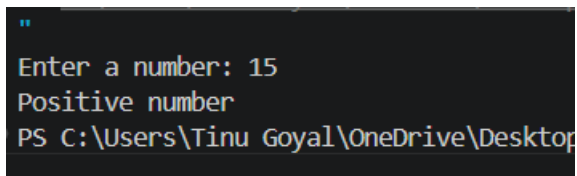
OUTPUT=

```
PS C:\Users\Tinu Goyal\OneDrive\Desktop\Tinu > "C:\Users\Tinu Goyal\AppData\Local\Programs\Python\Python113\python.exe" "C:\Users\Tinu Goyal\OneDrive\Desktop\Tinu\PROJECT2.py"
Enter first number: 3
Enter second number: 2
Sum: 5.0
Difference: -1.0
Product: 6.0
Quotient: 0.5
PS C:\Users\Tinu Goyal\OneDrive\Desktop\Tinu >
```

- ```
3. num = float(input("Enter a number: "))
```

```
if num > 0:
    print("Positive number")
elif num == 0:
    print("Zero")
else:
    print("Negative number")
```

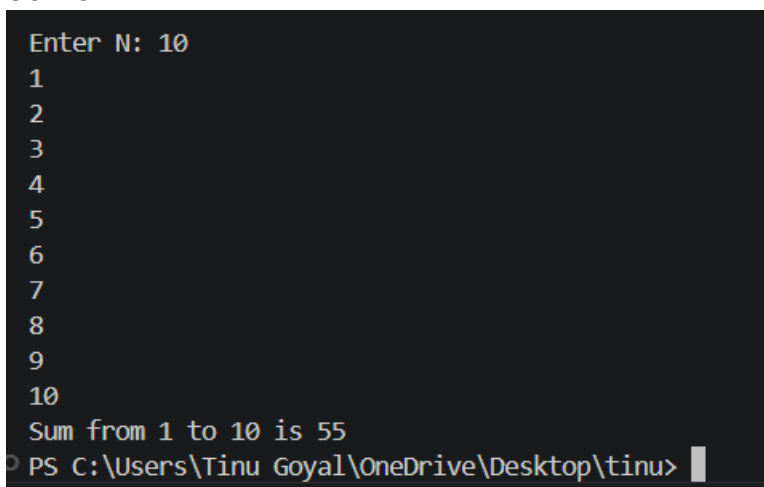
OUTPUT =



```
"  
Enter a number: 15  
Positive number  
PS C:\Users\Tinu Goyal\OneDrive\Desktop
```

4. `n = int(input("Enter N: "))`

```
total = 0  
for i in range(1, n + 1):  
    print(i)  
    total += i  
  
print("Sum from 1 to", n, "is", total)  
OUTPUT =
```

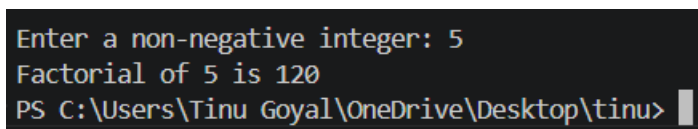


```
Enter N: 10  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
Sum from 1 to 10 is 55  
PS C:\Users\Tinu Goyal\OneDrive\Desktop\tinu>
```

5. `def factorial(n):`
 `if n == 0 or n == 1:`
 `return 1`
 `return n * factorial(n - 1)`

```
num = int(input("Enter a non-negative integer: "))  
print("Factorial of", num, "is", factorial(num))
```

OUTPUT =



```
Enter a non-negative integer: 5  
Factorial of 5 is 120  
PS C:\Users\Tinu Goyal\OneDrive\Desktop\tinu>
```

6. `numbers = [10, 20, 30, 40]`

```
print("First element:", numbers[0])  
numbers.append(50)    # add at end  
numbers[1] = 25       # modify
```

```
print("Updated list:", numbers)
print("Elements:")
for x in numbers:
    print(x)
```

OUTPUT=

```
First element: 10
Updated list: [10, 25, 30, 40, 50]
Elements:
10
25
30
40
50
```

```
7. s = input("Enter a string: ")
reversed_s = s[::-1]
print("Reversed string:", reversed_s)
vowels = "aeiouAEIOU"
count = 0
for ch in s:
    if ch in vowels:
        count += 1
```

```
print("Number of vowels:", count)
```

OUTPUT=

```
PS C:\Users\Tinu Goyal\OneDrive\Desktop\tinu> & "C:\Python39\python.exe" & "C:\Users\Tinu Goyal\OneDrive\Desktop\tinu\7.py"
Enter a string: fight
Reversed string: thgif
PS C:\Users\Tinu Goyal\OneDrive\Desktop\tinu>
```

```
8. def add(x, y): return x + y
def sub(x, y): return x - y
def mul(x, y): return x * y
def div(x, y): return x / y if y != 0 else float('inf')
```

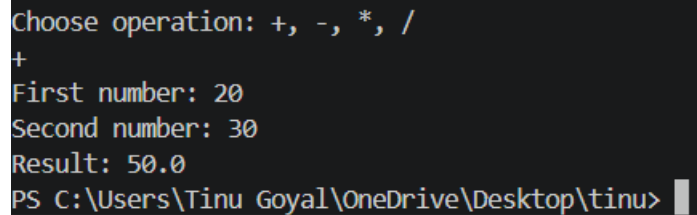
```
print("Choose operation: +, -, *, /")
op = input()
a = float(input("First number: "))
b = float(input("Second number: "))
```

```
if op == '+':
    print("Result:", add(a, b))
elif op == '-':
```

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```
        print("Result:", sub(a, b))
    elif op == '*':
        print("Result:", mul(a, b))
    elif op == '/':
        print("Result:", div(a, b))
    else:
        print("Invalid operator")
```

OUTPUT=



```
Choose operation: +, -, *, /
+
First number: 20
Second number: 30
Result: 50.0
PS C:\Users\Tinu Goyal\OneDrive\Desktop\tinu>
```

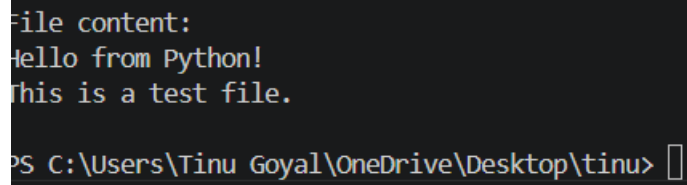
9. filename = "note.txt"

```
# Write to file
with open(filename, "w") as f:
    f.write("Hello from Python!\n")
    f.write("This is a test file.\n")
```

```
# Read from file
with open(filename, "r") as f:
    content = f.read()
```

```
print("File content:")
print(content)
```

OUTPUT=



```
File content:
Hello from Python!
This is a test file.
PS C:\Users\Tinu Goyal\OneDrive\Desktop\tinu>
```

10. import random

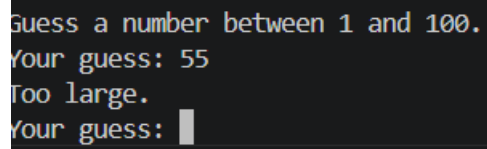
```
target = random.randint(1, 100)
print("Guess a number between 1 and 100.")
```

```
while True:
    guess = int(input("Your guess: "))
    if guess == target:
```

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```
        print("Correct! You won.")  
        break  
    elif guess < target:  
        print("Too small.")  
    else:  
        print("Too large.")
```

OUTPUT=

A terminal window with a dark background and light-colored text. The text shows the program's output: a prompt to guess a number between 1 and 100, followed by the user's input of 55, and the program's response "Too large." followed by a new prompt for a guess. A cursor is visible at the end of the second prompt.

```
Guess a number between 1 and 100.  
Your guess: 55  
Too large.  
Your guess: █
```